HARD · PROBLEM-SOLVING AND DATA ANALYSIS · ANSWER KEY

Mixed

01

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1 8

The correct answer is 8. The 6 students represent $(100-15-45-25)\% = 15\%$ of those invited to join the committee. If x people were invited to join the committee, then $0.15x = 6$. Thus, there were $\frac{6}{0.15} = 40$ people invited to join the committee. It follows that there were $0.45(40) = 18$ teachers and $0.25(40) = 10$ school and district administrators invited to join the committee. Therefore, there were 8 more teachers than school and district administrators invited to join the committee.

Q2 4176

The correct answer is 4,176. It's given that the side length of the larger square is 3 times the side length of the smaller square. This means that the area of the larger square is 3^2, or 9, times the area of the smaller square. If the area of the smaller square is represented by x, then the area of the larger square can be represented by $9x$. Therefore, the flat surface of the two adjacent squares has a total area of $x + 9x$, or $10x$. It's given that an electric field with strength 29.00 volts per meter passes uniformly through this surface and the total electric flux of the electric field through this surface is 4,640 volts · meters. Since it's given that the electric flux is the product of the electric field's strength and the area of the surface, the equation $29.0010x = 4{,}640$, or $290x = 4{,}640$, can be used to represent this situation. Dividing each side of this equation by 290 yields $x = 16$. Substituting 16 for x in the expression for the area of the larger square, $9x$, yields 916, or 144, square meters. Since the area of the larger square is 144 square meters, the electric flux, in volts · meters, of the electric field through the larger square can be determined by multiplying the area of the larger square by the strength of the electric field. Thus, the electric flux is $144 \text{ square meters} \frac{29.00 \text{ volts}}{\text{meter}}$, or 4,176 volts · meters.

Q3 182

The correct answer is 182. Let s represent the number of small candles the owner can purchase, and let l represent the number of large candles the owner can purchase. It's given that the owner pays $\$4.90$ per candle to purchase small candles and $\$11.60$ per candle to purchase large candles. Therefore, the owner pays $4.90s$ dollars for s small candles and $11.60l$ dollars for l large candles, which means the owner pays a total of $4.90s + 11.60l$ dollars to purchase candles. It's given that the owner budgets $\$2{,}200$ to purchase candles. Therefore, $4.90s + 11.60l \le 2{,}200$. It's also given that the owner must purchase a minimum of 200 candles. Therefore, $s + l \ge 200$. The inequalities $4.90s + 11.60l \le 2{,}200$ and $s + l \ge 200$ can be combined into one compound inequality by rewriting the second inequality so that its left-hand side is equivalent to the left-hand side of the first inequality. Subtracting l from both sides of the inequality $s + l \ge 200$ yields $s \ge 200 - l$. Multiplying both sides of this inequality by 4.90 yields $4.90s \ge 4.90200 - l$, or $4.90s \ge 980 - 4.90l$. Adding $11.60l$ to both sides of this inequality yields $4.90s + 11.60l \ge 980 - 4.90l + 11.60l$, or $4.90s + 11.60l \ge 980 + 6.70l$. This inequality can be combined with the inequality $4.90s + 11.60l \le 2{,}200$, which yields the compound inequality $980 + 6.70l \le 4.90s + 11.60l \le 2{,}200$. It follows that $980 + 6.70l \le 2{,}200$. Subtracting 980 from both sides of this inequality yields $6.70l \le 2{,}200$. Dividing both sides of this inequality by 6.70 yields approximately $l \le 182.09$. Since the number of large candles the owner purchases must be a whole number, the maximum number of large candles the owner can purchase is the largest whole number less than 182.09, which is 182.

Q4

A

A. $y = 215x + 215$

Choice A is correct. It's given that the cost of renting a large canopy tent is $\$430$ for the first day and $\$215$ for each additional day for up to 10 days. For x days of renting the tent, the cost includes $\$430$ for the first day and $\$215$ for each of the $(x - 1)$ additional days. It follows that the cost y, in dollars, of renting the tent can be expressed as $y = 430 + 215(x - 1)$, which is equivalent to $y = 430 + 215x - 215$, or $y = 215x + 215$. Therefore, the equation $y = 215x + 215$ gives the cost of renting the tent for x days, where x is a positive integer and $x \leq 10$.

Choice B is incorrect. This equation represents a situation where the cost of renting the tent for the first day is $\$215$, not $\$430$, and the cost for each additional day is $\$430$, not $\$215$.

Choice C is incorrect. This equation represents a situation where the cost of renting the tent for the first day is $\$645$, not $\$430$, and the cost for each additional day is $\$430$, not $\$215$.

Choice D is incorrect. This equation represents a situation where the cost of renting the tent for the first day is $\$645$, not $\$430$.

Q5

87

The correct answer is 87. It's given that in August, the car dealer completed 15 more than 3 times the number of sales the car dealer completed in September. Let x represent the number of sales the car dealer completed in September. It follows that $3x + 15$ represents the number of sales the car dealer completed in August. It's also given that in August and September, the car dealer completed 363 sales. It follows that $x + 3x + 15 = 363$, or $4x + 15 = 363$. Subtracting 15 from each side of this equation yields $4x = 348$. Dividing each side of this equation by 4 yields $x = 87$. Therefore, the car dealer completed 87 sales in September.

Q6

24.5

Accept also: 49/2

The correct answer is 24.5. It's given that a triangle is formed by connecting the three points shown, which are $-3, 4$, $5, 3$, and $4, -3$. Let this triangle be triangle A. The area of triangle A can be found by calculating the area of the rectangle that circumscribes it and subtracting the areas of the three triangles that are inside the rectangle but outside triangle A. The rectangle formed by the points $-3, 4$, $5, 4$, $5, -3$, and $-3, -3$ circumscribes triangle A. The width, in units, of this rectangle can be found by calculating the distance between the points $5, 4$ and $5, -3$. This distance is $4 - -3$, or 7. The length, in units, of this rectangle can be found by calculating the distance between the points $5, 4$ and $-3, 4$. This distance is $5 - -3$, or 8. It follows that the area, in square units, of the rectangle is 78, or 56. One of the triangles that lies inside the rectangle but outside triangle A is formed by the points $-3, 4$, $5, 4$, and $5, 3$. The length, in units, of a base of this triangle can be found by calculating the distance between the points $5, 4$ and $5, 3$. This distance is $4 - 3$, or 1. The corresponding height, in units, of this triangle can be found by calculating the distance between the points $5, 4$ and $-3, 4$. This distance is $5 - -3$, or 8. It follows that the area, in square units, of this triangle is $\frac{1}{2}81$, or 4. A second triangle that lies inside the rectangle but outside triangle A is formed by the points $4, -3$, $5, 3$, and $5, -3$. The length, in units, of a base of this triangle can be found by calculating the distance between the points $5, 3$ and $5, -3$. This distance is $3 - -3$, or 6. The corresponding height, in units, of this triangle can be found by calculating the distance between the points $5, -3$ and $4, -3$. This distance is $5 - 4$, or 1. It follows that the area, in square units, of this triangle is $\frac{1}{2}16$, or 3. The third triangle that lies inside the rectangle but outside triangle A is formed by the points $-3, 4$, $-3, -3$, and $4, -3$. The length, in units, of a base of this triangle can be found by calculating the distance between the points $4, -3$ and $-3, -3$. This distance is $4 - -3$, or 7. The corresponding height, in units, of this triangle can be found by calculating the distance between the points $-3, 4$ and $-3, -3$. This distance is $4 - -3$, or 7. It follows that the area, in square units, of this triangle is $\frac{1}{2}77$, or 24.5. Thus, the area, in square units, of the triangle formed by connecting the three points shown is $56 - 4 - 3 - 24.5$, or 24.5. Note that 24.5 and 49/2 are examples of ways to enter a correct answer.

Q7

.3928

Accept also: `.3929`, `11/28`

The correct answer is $\frac{11}{28}$. The cosine of an acute angle in a right triangle is defined as the ratio of the length of the leg adjacent to the angle to the length of the hypotenuse. In the triangle shown, the length of the leg adjacent to the angle with measure $x°$ is 11 units and the length of the hypotenuse is 28 units. Therefore, the value of $\cos x°$ is $\frac{11}{28}$. Note that 11/28, .3928, .3929, 0.392, and 0.393 are examples of ways to enter a correct answer.

Q8

D

D. The estimated number of advertisements the company sent to its clients in 1997 was $9{,}000$.

Choice D is correct. The y-intercept of a graph in the xy-plane is the point where $x = 0$. For the given function f, the y-intercept of the graph of $y = fx$ in the xy-plane can be found by substituting 0 for x in the equation $y = 9{,}0000.66^x$, which gives $y = 9{,}0000.66^0$. This is equivalent to $y = 9{,}0001$, or $y = 9{,}000$. Therefore, the y-intercept of the graph of $y = fx$ is $0, 9{,}000$. It’s given that the function f models the number of advertisements a company sent to its clients each year. Therefore, fx represents the estimated number of advertisements the company sent to its clients each year. It's also given that x represents the number of years since 1997. Therefore, $x = 0$ represents 0 years since 1997, or 1997. Thus, the best interpretation of the y-intercept of the graph of $y = fx$ is that the estimated number of advertisements the company sent to its clients in 1997 was $9{,}000$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q9

B

B. -50

Choice B is correct. It's given that the equation $-2x^2 + 20x + c = 0$, where c is a constant, has exactly one solution. A quadratic equation of the form $ax^2 + bx + c = 0$ has exactly one solution if and only if its discriminant, $b^2 - 4ac$, is equal to zero. It follows that for the given equation, $a = -2$ and $b = 20$. Substituting -2 for a and 20 for b in $b^2 - 4ac$ yields $20^2 - 4(-2)(c)$, or $400 + 8c$. Since the discriminant must equal zero, it follows that $400 + 8c = 0$. Subtracting 400 from both sides of this equation yields $8c = -400$. Dividing each side of this equation by 8 yields $c = -50$. Therefore, the value of c is -50.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10

361/8

Accept also: `45.12`, `45.13`

The correct answer is $\frac{361}{8}$. The rational exponent property is $\sqrt[n]{y^m} = y^{\frac{m}{n}}$, where $y > 0$, m and n are integers, and $n > 0$. This property can be applied to rewrite the given expression $6\sqrt[5]{3^5x^{45}} \cdot \sqrt[8]{2^8x}$ as $63^{\frac{5}{5}}x^{\frac{45}{5}}2^{\frac{8}{8}}x^{\frac{1}{8}}$, or $63x^9 2x^{\frac{1}{8}}$. This expression can be rewritten by multiplying the constants, which gives $36x^9x^{\frac{1}{8}}$. The multiplication exponent property is $y^n \cdot y^m = y^{n+m}$, where $y > 0$. This property can be applied to rewrite the expression $36x^9x^{\frac{1}{8}}$ as $36x^{9+\frac{1}{8}}$, or $36x^{\frac{73}{8}}$. Therefore, $6\sqrt[5]{3^5x^{45}} \cdot \sqrt[8]{2^8x} = 36x^{\frac{73}{8}}$. It's given that $6\sqrt[5]{3^5x^{45}} \cdot \sqrt[8]{2^8x}$ is equivalent to ax^b; therefore, $a = 36$ and $b = \frac{73}{8}$. It follows that $a + b = 36 + \frac{73}{8}$. Finding a common denominator on the right-hand side of this equation gives $a + b = \frac{288}{8} + \frac{73}{8}$, or $a + b = \frac{361}{8}$. Note that 361/8, 45.12, and 45.13 are examples of ways to enter a correct answer.

Notes

Accept equivalent forms (e.g. `0.25` ≡ `1/4`). For grid-in items, decimal and fraction answers are both valid.

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